

EECE 571Q

Lecture 04: Just stay still!

Thursday 21 May 2026

Announcements

- Classes on Zoom next week - link in Piazza
- Quiz 2 on Tuesday

Last time

We defined conditions for quantum error correction

“($Q, \mathcal{E}$) is a QECC iff $\forall |\psi\rangle, |\phi\rangle \in Q, \forall E_a, E_b \in \mathcal{E}$,

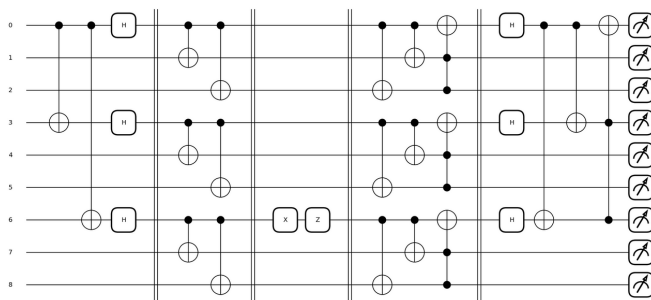
$$\langle \psi | E_a^\dagger E_b | \phi \rangle = C_{ab} \langle \psi | \phi \rangle,$$

Note that C_{ab} does not depend on $|\psi\rangle$ or $|\phi\rangle$ ”

Source: Gottesman, Thm. 2.7

Last time

We combined the bit and phase flip codes into a 9-qubit code that can correct one arbitrary single-qubit error.



Learning outcomes

- 1 Express quantum error correcting codes in the stabilizer formalism
- 2 Characterize whether Pauli errors are correctable by a code using its stabilizer generators
- 3 Define classical linear codes, and formulate them in terms of generator and parity check matrices
- 4 Use classical linear codes to construct CSS codes for quantum error correction

Stabilizers

$$\mathcal{P}_n = \{I, X, Y, Z\}^{\otimes n} \times \{1, -1, i, -i\}$$

Let T be a QECC. The **stabilizer** of the code, $S(T)$, is $|\mathcal{P}_n| = 4^{n+1}$

$$S(T) = \left\{ M \in \mathcal{P}_n \mid M|\psi\rangle = |\psi\rangle \right. \\ \left. \forall |\psi\rangle \in T \right\} \quad \mathcal{P}_n = \{I, X, Y, Z\}^{\otimes n}$$

In words: a (sub)group of Paulis for which all codewords are $+1$ eigenstates.

Facts about $S(T)$:

- $-I$ is never in $S(T)$
- all elements of $S(T)$ commute (Abelian subgroup of $\mathcal{P}_n$)
- choosing a subgroup S defines the stabilized codewords of T

Stabilizer codes

More convenient to work “backwards”: given S , define codespace

$$\mathcal{T}(S) = \left\{ |\psi\rangle \text{ s.t. } M|\psi\rangle = |\psi\rangle \quad \forall M \in S \right\}$$

If $T = \mathcal{T}(S(T))$, T is a **stabilizer code**.

Bit flip code revisited

An alternative way of thinking about the parity computation is in terms of eigenvalues w.r.t. the code's stabilizer group.

Consider our logical states:

$$|\bar{0}\rangle = |000\rangle$$

$$P|000\rangle = |000\rangle$$

$$|\bar{1}\rangle = |111\rangle$$

$$P|111\rangle = |111\rangle$$

Which three-qubit Paulis are these states +1 eigenstates of?

Z s on any qubit of $|000\rangle \rightarrow |000\rangle \checkmark$

$$\left. \begin{array}{l} ZZI \\ ZIZ \\ IZZ \end{array} \right\} S$$

$$\begin{aligned} (ZZI) (\alpha|000\rangle + \beta|111\rangle) \\ = \alpha|000\rangle + \beta|111\rangle \end{aligned}$$

Bit flip code revisited

Let's determine a *minimal* representation of the group in terms of its **stabilizer generators**.

$$\begin{array}{c} ZZI \\ IZZ \\ \hline ZIZ \end{array}$$

$$S = \langle ZZI, IZZ \rangle$$

$$\begin{array}{l} P_1, P_2, \dots, P_r \quad \longrightarrow \quad P_1, P_2, P_3, P_4, \dots \quad \cdot \text{fix order} \\ P^2 = I \quad \quad \quad P_1^{b_1}, P_2^{b_2}, \dots, P_r^{b_r} \quad b_i \in \{0, 1\} \end{array}$$

Exercise: if there are r stabilizer generators, how large is the stabilizer group S ? $|S| = 2^r$ (incl. identity)

Stabilizers generators

We can also work backwards to obtain codewords from the stabilizer generators by defining a projector onto the codespace,

$$\Pi_S = \frac{1}{2^r} \prod_{i=1}^r (I + M_i)$$

Π_S provides insight into size of the stabilized space: each generator cuts space in half, so r generators gives space with dimension 2^{n-r} .

Special case: if $r = n$, have a single state called a *stabilizer state*.

$$|\psi\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle) \Rightarrow S = \{ZZ, XX\}$$

Bit flip code: stabilizers and logical operations

$$\langle ZZI, IZZ \rangle$$

Logical Z: need a Pauli that

- ~~stabilizes the logical state~~ ^{keeps us in code space} but is *not* in S
- *commutes* with everything in S
- has the right action on the subspace

$$\begin{aligned}\bar{Z} |0\rangle &= |0\rangle \\ \bar{Z} |1\rangle &= -|1\rangle\end{aligned}$$

$$\Rightarrow \bar{Z} = ZZZ$$

Logical X: need a Pauli that

- ~~stabilizes the logical state~~ ^{keeps us in code space} but is *not* in S
- *commutes* with everything in S
- has the right action on the subspace
- anticommutes with logical Z

$$\begin{aligned}\bar{X} |0\rangle &= |1\rangle \\ \bar{X} |1\rangle &= |0\rangle\end{aligned}$$

$$\bar{Z}\bar{X} = -\bar{X}\bar{Z} \Rightarrow XXX = \bar{X}$$

Phase flip code: stabilizer formalism

$$|0\rangle = |+++ \rangle$$

$$|1\rangle = |-- \rangle$$



$$XXI, XI\bar{X}, I\bar{X}\bar{X}$$



$$S = \langle XXI, I\bar{X}\bar{X} \rangle$$

$$\bar{Z}|0\rangle = |0\rangle$$

$$\bar{Z}|1\rangle = -|1\rangle$$

$$\bar{X}|0\rangle = |1\rangle$$

$$\bar{X}|1\rangle = |0\rangle$$

$$\bar{Z} = XII, (XXX, I\bar{X}\bar{X}, \dots)$$

$$\bar{X} = ZZZ$$

Stabilizer measurement and error detection

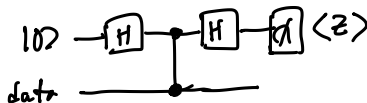
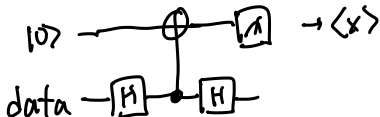
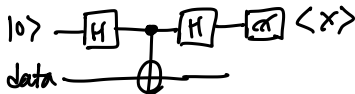
We can use this formalism to construct circuits for error detection and recovery: simply *measure the stabilizer generators*.



$M: \pm 1$ eig. vals.



$\Rightarrow$



Correctable errors anticommute with at least one generator. We can detect their presence in the syndrome measurement.

Bit flip

XII

$\langle ZZII, IZZ \rangle$

anticomm.

-1 expval ZZI
 $+1$ expval IZZ

Stabilizers and logical errors

Danger: error that *commutes* with all elements of S , but isn't in S .

More precisely, define the *normalizer* $N(S)$ as

$$N(S) = \{ N \in \mathcal{P}_n \mid [M, N] = 0 \ \forall M \in S \}$$

For an arbitrary error E ...

- if $E \notin N(S)$, E is detectable
- if $E \in N(S)$ and $E \in S$, the code space is unchanged
- if $E \in N(S)$ but $E \notin S$, E is **undetectable**

$$S = (ZZI, IZZ, ZIZ) \quad ZII \text{ ☹}$$

Stabilizers and logical errors

Gottesman Thm 3.4

“The set of undetectable errors for a stabilizer code S is $\hat{N}(S) \setminus \hat{S}$. The **distance** of S is $\min\{\text{wt}(E) \mid \hat{E} \in \hat{N}(S) \setminus \hat{S}\}$ ”.

Gottesman Thm 3.5

“The stabilizer code S corrects a set of errors $\mathcal{E} \subseteq \mathcal{P}_n$ iff $\hat{E}^\dagger \hat{F} \notin \hat{N}(S) \setminus \hat{S}$ for all $E, F \in \mathcal{E}$ ”.

Shor code: stabilizer formalism

Stabilizer code usually described by notation $[[n, k, d]]$. The 9-qubit Shor code is a $[[9, 1, 3]]$ code.

Name	Operator
<i>gens.</i> g_1	$ZZIIIIIIII$
g_2	$IIZZIIIIII$
g_3	$IIIZZIIII$
g_4	$IIIIIZZIII$
g_5	$IIIIIIIZZ I$
g_6	$IIIIIIII ZZ$
g_7	$XXXXIXXI$
g_8	$IIIXXXXXX$
$\bar{Z}$	$XXXXXXXXXX$
$\bar{X}$	$ZZZZZZZZZZ$

logical ops

IIIZZIIIIII

What is an example of a weight-3 undetectable error?

Screenshot: Nielsen & Chuang, chapter 10.5.6.

Syndromes and decoding

Exercise: What are the error syndromes for Y_1 and $X_0Z_1X_2$?

Name	Operator	Y_1	$X_0Z_1X_2$
g_1	$ZZIIIIIIII$	-1	-1
g_2	$IZZIIIIIIII$	-1	-1
g_3	$IIIZZIIIIII$		
g_4	$IIIIIZZIIII$		
g_5	$IIIIIIIZZII$		
g_6	$IIIIIIIIIZZ$		
g_7	$XXXXXXIIII$	-1	-1
g_8	$IIIXXXXXXX$		
$\bar{Z}$	$XXXXXXXXXX$		
$\bar{X}$	$ZZZZZZZZZZ$		

Screenshot: Nielsen & Chuang, chapter 10.5.6.

Syndromes and decoding

Two errors, E and F are in the same **coset** of $N(S)$ if $F = EN$ for some $N \in N(S)$. Errors in the same coset of $N(S)$ have the same error syndrome.

We will explore this more when we discuss decoding.

Where do codes come from?

We have seen the bit flip, phase flip codes, and 9-qubit Shor code. They are all related.

*bit / Phase flip codes: same codes
different bases*

Shor code: concatenates the two.

- Are there better codes?
- Are there smaller codes?
- How do we find new ones?

Classical codes

Many QECCs are constructed from *classical* error correcting codes.

Source: Gottesman, Def. 4.1

"A classical error-correcting code $(e, \mathcal{E})$ is a **map** $e : [1 \dots K] \rightarrow [1 \dots M]$ with the set of **correctable errors** $\mathcal{E}$ (consisting of functions $E : [1 \dots M] \rightarrow [1 \dots M]$) with the following property: $\exists$ function $d : [1 \dots M] \rightarrow [1 \dots K]$ such that $\forall E \in \mathcal{E}, \forall x \in [1 \dots K],$ "

$$d(E(e(x))) = x$$

$$E(e(x)) = F(e(y)) \quad \text{not correctable}$$

Classical codes

The *distance* of a code $C \subseteq \mathbb{Z}_N$ is

$$N = 2^n \text{ or } q^n$$

$$\min(wt(E) \mid \exists x \neq y \in C, Ex = y)$$

$$\begin{matrix} 000 \\ 111 \end{matrix} \rightarrow E = (111)$$

Consequence: a distance d code

- corrects general errors on up to $\lfloor (d-1)/2 \rfloor$ coordinates
- correct erasure errors on up to $d-1$ coordinates
- detect general errors on up to $d-1$ coordinates

$$\begin{matrix} 00000 \\ 11111 \\ \dots \end{matrix}$$

Notation: (n, K, d) code

$$\begin{matrix} 00011 \\ 11111 \\ 01111 \end{matrix}$$

Classical linear codes

Source: Gottesman, Def. 4.3, section 4.2

A code $C \subseteq \mathbb{Z}_2^n$ is a *linear code* if

$$x, y \in C \rightarrow x + y \in C$$

Codewords are linear combinations of basis states for (sub)space C .

Example: repetition code. $k = 1$ basis state, $2^k = 2$ codewords.

$$x = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$0x = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$1 \cdot x = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

Classical linear codes

Given k basis vectors $x_1, \dots, x_k$, define a $k \times n$ **generator matrix**

$$G_C = \begin{pmatrix} x_1 \\ \vdots \\ x_k \\ \vdots \\ x_n \end{pmatrix}$$

The linear map $v \in \mathbb{Z}_2^k \rightarrow G_C^T v \in \mathbb{Z}_2^n$ is the *encoder* for C .

Example: repetition code.

$$|0\rangle \rightarrow \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad |1\rangle \rightarrow \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$G_C = (1 \quad 1 \quad 1) \quad G_C^T |0\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad G_C^T |1\rangle = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

Classical linear codes

Given G_C , a $(n - k) \times n$ **parity check matrix** H_C contains rows y_i such that $G y_i = 0$, with $\{y_i\}$ a maximal linearly independent set.

Relationship between G_C and H_C : $G_C H_C^T = H_C G_C^T = 0$

Example: repetition code.

$$H = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$$

$$H_C G_C^T = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \pmod{2}$$

Classical linear codes

How to model errors?

$$x \rightarrow x + e$$

Apply the parity check matrix...

$$H_c (x + e) = \underbrace{H_c x}_0 + H_c e = \underline{\underline{H_c e}}$$

Example: repetition code.

$$x = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad e = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$$H_c e = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$


Classical codes

(Source: Gottesman Ch. 4.2) *We will start here Tuesday.*

The *distance* of a code $C \subseteq \mathbb{Z}_N$ is

$$\min\{\text{wt}(E) \mid \exists x \neq y \in C, Ex = y\}$$

The *distance* of a linear code C is

A linear code C corrects error set $\mathcal{E}$

■ iff

■

The Hamming code

Suppose we wish to correct a 1-bit error on r -bit code words. Need all error syndromes to be unique.

Code parameters:

General parameters:

Dual codes

Given code C , its dual code $C^\perp$ is

The dual code exchanges roles of G and H .

If C has code parameters $[n, k, d]$, $C^\perp$ has parameters $[n, n - k, d]$.

Relationship with stabilizer codes

Binary symplectic representation: isomorphism between $\hat{\mathcal{P}}_n$ and vectors in $\mathbb{Z}_2^n \times \mathbb{Z}_2^n$.

Commutator computed from symplectic inner product:

We can represent stabilizers in this format:

Relationship with stabilizer codes

(Source: Gottesman Thm. 4.4)

More generally, given a linear code C and parity check matrix H , define stabilizer code with binary symplectic representation,

This stabilizer code has the same basis codewords as C (plus linear combinations thereof), and corrects the same set of errors.

CSS codes

(Source: Gottesman Def. 5.1)

A stabilizer code is a CSS code if there exists a choice of stabilizer generators with binary symplectic representation of the form

Example: choose both A and B to be H of $[7, 4, 3]$ Hamming code:

The Steane code

Stabilizer generators:

What are the logical operations? What are the code parameters?

Relationship with stabilizer codes

(Source: Gottesman Thm. 5.1)

Let C_1 be an $[n, k_1, d_1]$ classical linear code with parity check matrix H_1 , and C_2 be an $[n, k_2, d_2]$ classical linear code with parity check matrix H_2 , with $C_1^\perp \subseteq C_2$.

The stabilizer code (CSS) code with stabilizer generators

is an $[[n, k, d]]$ quantum code with $k = k_1 + k_2 - n$ and $d \geq \min(d_1, d_2)$.

Next time

Next class:

- Some bounds and existence criteria for QECCs
- The perfect code ($[[5, 1, 3]]$)

Action items:

- ➊ Prepare for quiz 2
- ➋ Watch PrairieLearn for project information

Recommended reading:

- From this class: Gottesman Ch. 3-4; N&C 10.5
- For next class: Gottesman Ch. 3-4; N&C 10.1-10.5